Society of Computer Science and Engineering

XAVENIR – 2026 | AI/ML Challenge

RULES:

1. Problem Statement 1 is open to all students including 1st years.
2. Problem Statement 2 is only for 1st-year B.Tech. students.
3. Problem Statement 1 is eligible for overall prizes.
4. Problem Statement 2 has a separate exclusive prize for 1st-year participants.
5. Participants are free to use any resources, datasets, or tools —

   EXCEPT pretrained model backbones or pretrained embeddings of any kind.

   (No BERT, RoBERTa, GPT, Word2Vec, GloVe, FastText, or any pretrained weights. All representations must be learned from scratch.) 6. Final submission must include the complete code and a PPT covering:

   Approach, Methodology, Results, and Evaluation Metrics.

PROBLEM STATEMENT 1 (Open to All Years):

"Catch the Liar" — Hallucination Detection in AI-Generated Text

Background:

Large Language Models are increasingly used to summarize scientific content. However, these models can be manipulated — causing them to hallucinate, drift off-topic, introduce factual errors, or produce outputs subtly inconsistent with the source. In this challenge, you are given pairs of AI-generated summaries of the same article. One is from a clean model, the other from a manipulated one. Your job is to find the liar.

What You Need to Do:

Given a pair of summaries (Text A, Text B) of the same source article, output:

- A label: 1 if Text A is the hallucinated output, 2 if Text B is.

No pretrained backbones. Every feature and representation must be built from scratch on the provided data.

Approaches to Explore:

- Linguistic features: vocabulary richness, sentence length variance, POS tag ratios, readability scores, Unicode character ratios.
- Statistical divergence between the two texts (e.g., KL divergence over word distributions).
- Train your own tokenizer and embeddings from scratch, then build a simple classifier (Naive Bayes, SVM, shallow MLP).
- Consistency checks: does the summary stay on topic? Are named entities from the source preserved?

Dataset:

- ESO "The Messenger" journal articles + LLM-generated summary pairs.
- https://www.kaggle.com/competitions/fake-or-real-the-impostor-hunt/data
- Training set has ground truth labels. Test set does not.

Evaluation Criteria:

- Accuracy on test set (primary metric)
- Novelty and clarity of approach

• Simplicity vs. performance tradeoff

References:

[1] Kaczmarek et al., "Fake or Real: The Impostor Hunt in Texts for Space Operations", arXiv:2507.13508v3, 2025.

[2] Sriramanan et al., "LLM-Check: Investigating Detection of Hallucinations in Large Language Models", NeurIPS 2024, pp. 34188-34216.

[3] Yehuda et al., "InterrogateLLM: Zero-Resource Hallucination Detection in LLM-Generated Answers", 2024.